EXPERIMENT 8

PIGLATINLANGUAGE-PIG

OBJECTIVE:

1. InstallationofPIG.

RESOURCES:

VMWare,Webbrowser,4GBRAM, HardDisk80GB.

PROGRAMLOGIC:

STEPSFORINSTALLINGAPACHE PIG

1) Extractthepig-0.15.0.tar.gzandmovetohomedirectory

2) SettheenvironmentofPIGinbashrcfile.

3) Pig can run in two modes

LocalModeandHadoopMode

Pig –x local and pig

4) GruntShell

Grunt >

5) LOADINGDataintoGruntShell

DATA=LOAD<CLASSPATH>USINGPigStorage(DELIMITER) as(ATTRIBUTE:

DataType1,ATTRIBUTE: DataType2…..)

6) DescribeData

DescribeDATA;

7) DUMPData

Dump DATA;

INPUT/OUTPUT:

InputasWebsiteClickCountData

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6) DescribeData

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7) DUMPData

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INPUT/OUTPUT:

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```
grunt> ad1 = load '/home/lendi/Desktop/static_data/ad_data/ad_data1.txt' using P
igstorage(',') as (item:chararray,campaignId:chararray,date:chararray,time:char
array,display_site:chararray,was_clicked:int,cpc:int,country:chararray,placement
:chararray);
2016-10-14 02:35:32,441 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation : io.bytes.per.checksum is deprecated. Instead, use dfs.bytes-per-checksum
2016-10-14 02:35:32,441 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation : fs.default.name is deprecated. Instead, use fs.defaultFS
grunt> describe ad1;
ad1: {item: chararray,campaignId: chararray,date: chararray,time: chararray,disp
lay_site: chararray,was_clicked: int,cpc: int,country: chararray,placement: char
array}
grunt> ad2 = load '/home/lendi/Desktop/static_data/ad_data/ad_data2.txt' using P
igstorage(',') as (campaignId:chararray,date:chararray,time:chararray,display_si
te:chararray,placement:chararray,was_clicked:int,cpc:int,item:chararray);
2016-10-14 02:36:08,732 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation : io.bytes.per.checksum is deprecated. Instead, use dfs.bytes-per-checksum
2016-10-14 02:36:08,732 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation : fs.default.name is deprecated. Instead, use fs.defaultFS
grunt> describe ad2;
ad2: {campaignId: chararray,date: chararray,time: chararray,display_site: charar
ray,placement: chararray,was_clicked: int,cpc: int,item: chararray}
grunt>
```

PRE-LABVIVAQUESTIONS:

1) Whatdoyoumeanbyabaginpig?
2) DifferentiatebetweenPigLatinandHiveQL
3) Howwillyou mergethecontentsoftwoormorerelationsanddividearanglerelation into two or more relations?

LAB ASSIGNMENT:

1. ProcessbaseballdatausingApachePig

POST-LABVIVA QUESTIONS:

1. Whatistheusageofforeachoperationinpigscripts?
2. WhatdoesFlattendoinPig

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EXPERIMENT -9

PIGCOMMANDS

OBJECTIVE:

WritePigLatinscriptssort,group,join, project,andfilteryourdata.

RESOURCES:

VMWare, Webbrowser,4GBRAM,HardDisk80GB.

PROGRAMLOGIC:

FILTER Data

FDATA= FILTERDATAby ATTRIBUTE=VALUE;

GROUPData

GDATA= GROUP DATAbyATTRIBUTE;

IteratingData

FOR_DATA=FOREACHDATAGENERATEGROUPASGROUP_FUN,

ATTRIBUTE =<VALUE>

SortingData

SORT_DATA=ORDERDATABYATTRIBUTEWITHCONDITION;

LIMITData

LIMIT_DATA=LIMIT DATACOUNT;

JOINData

JOINDATA1BY(ATTRIBUTE1,ATTRIBUTE2….),DATA2BY (ATTRIBUTE3,ATTRIBUTE….N)

INPUT/OUTPUT:

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```
lendi@ubuntu: ~

grunt> join_data = join ad1 by (campaignId,display_site,cpc),ad2 by (campaignId,
display_site,cpc);
grunt> describe join_data;
join_data: {ad1::item: chararray,ad1::campaignId: chararray,ad1::date: chararray
,ad1::time: chararray,ad1::display_site: chararray,ad1::was_clicked: int,ad1::cp
c: int,ad1::country: chararray,ad1::placement: chararray,ad2::campaignId: charar
ray,ad2::date: chararray,ad2::time: chararray,ad2::display_site: chararray,ad2::
placement: chararray,ad2::was_clicked: int,ad2::cpc: int,ad2::item: chararray}
grunt>
```


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PRE-LABVIVAQUESTIONS:

1. Howwillyou mergethecontentsoftwoormorerelationsanddivideasinglerelation into two or more relations?
2. WhatistheusageofforeachoperationinPigscripts?
3. WhatdoesFlattendoinPig?

LAB ASSIGNMENT:

1. UsingApachePigtodevelopUserDefinedFunctionsforstudentdata.

PRE-LABVIVAQUESTIONS:

1. Whatdoyoumeanbyabaginpig?
2. DifferentiatebetweenPigLatinandHiveQL

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